

Practice Worksheet 4: Iteration

CS 101 Algorithmic Problem Solving

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1. Facht Auriel's Riddle

You've bought an entrance ticket to Facht Auriel's Castle and your ticket number is N . The castle houses an ancient blade of immense power that can only be obtained by those wise enough to crack the riddles created by Facht Auriel himself. An entrance ticket gives you one shot at the riddles. However, nobody has been able to solve any of the riddles yet.

Finally, your turn to enter the castle arrives. You enter a large room in the center of which is a statue of Facht Auriel, with his two hands extended out. On one of his hands, there's a pen, a paper, and a riddle that says:

"You are the product of those that came before you and failed. Scribble who you are on this parchment and await Facht Auriel's judgement."

Confused by the riddle at first, you sit down to think. After some time, you realize the answer must be the product of the ticket numbers of all those who came before you.

Given your ticket number N , determine the answer to the riddle. Note that the ticket numbers start from 1.

HINT: USE "in" operator

Constraints

- $N \in \mathbb{N}$
- $2 \leq N \leq 10^5$

Interaction

The input comprises of a single integer denoting the value of N .

The output must be a single integer denoting the answer to the riddle.

Sample

Input	Output
4	6
3	2

In the first case, $N = 4$, so the answer to the riddle is $1 * 2 * 3 = 6$.

In the second case, $N = 3$, so the answer to the riddle is $1 * 2 = 2$.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
12	39916800
9	40320
6	120

Pseudocode

Input: N

Output:

a=1

n=8

for i in range(1,n):

a = a * i

print(a)

Dry Run

Using any one of the inputs provided in the Exercise section above, dry run your pseudocode in the space below.

Input: 6

a=1

n=6

for i in range(1,6):

a = a * 1 (i=1)

for i in range(1,6):

a = 1 * 2 (i=2)

for i in range(1,6):

a = 2 * 3 (i=3)

for i in range(1,6):

a = 6 * 4 (i=4)

for i in range(1,6):

a = 24 * 5 (a=120)

print(a)

Output: 120

2. Digit counter!

Ever found yourself curious about the number of digits hiding within a mysterious number? Instead of painstakingly counting them one by one, you're about to embark on a fascinating journey to create an algorithm that will seamlessly unveil the secret, making it a breeze to determine the number of digits in any given number.

Constraints

- $1 \leq N \leq 10^8$

Interaction

The input contains a single integer value denoting the number whose digits are to be counted.

The output would contain the number of digits present in the integer.

Sample

Input	Output
19	2
23871	5

In the first case, $N = 19$, there are two digits meaning the answer is 2.

In the second case, $N = 23871$, there are five digits meaning the answer is 5.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
28728172	8
48271234	8

Pseudocode

Input: N

Output:

a=0

while N>0:

 N=N//10

 a+=1

print(a)

Dry Run

Using any one of the inputs provided in the Exercise section above, dry run your pseudocode in the space below.

Input: 28728172

a=0

while 28728172>0: (true)

 N=28728172//10

 a+=1 (a=1)

while 2872817>0: (true)

 N=2872817//10

 a+=1 (a=2)

while 287281>0: (true)

 N=287281//10

 a+=1 (a=3)

while 28728>0: (true)

 N=28728//10

 a+=1 (a=4)

while 2872>0: (true)

 N=2872//10

 a+=1 (a=5)

while 287>0: (true)

 N=287//10

 a+=1 (a=6)

while 28>0: (true)

 N=28//10

 a+=1 (a=7)

while 2>0: (true)

 N=2//10

 a+=1 (a=8)

while 0>0: (False)

 (a=8)

print(a)

Output:8

3. Frustrated Teacher

A teacher frustrated with the lack of discipline of his students, decides to cancel class if fewer than some number of students are present when class starts. Arrival times t are in range from on time ($t \geq 0$) to arrived late ($t < 0$).

Given the total number of students n , arrival time of each student t and a threshold number of attendees a , check whether the teacher cancels the class or not.

Constraints

- $0 \leq n, a \leq 200$
- $-10 \leq t \leq 10$

Interaction

The input comprises a line containing 2 space-separated integers denoting the values of n and a respectively. Followed by a line containing n space separated integers each denoting the arrival time t of each student.

The output must contain Yes or No, stating whether the class is cancelled or not.

Sample

Input	Output
5 3	Yes
-1 0 2 -2 -4	
4 2	No
1 0 2 -1	

In the first case, only 2 students arrive on time while the threshold is 3. Therefore, the class is cancelled.

In the second case, 3 students arrive on time while the teacher required atleast 2 to arrive on time. Therefore, the class is not cancelled.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
6 4	No
-1 0 2 -2 4 1	
5 2	Yes
-1 0 -2 -1 -3	

Pseudocode

```

Input:
a, n,
Output:
n=int(input("enter total number of students: "))
b=0
a=int(input("enter how mant students are required on time"))
for i in range(1,n+1):
    t=int(input("enter time of student"))
    if t>=0:
        b+=1
    else:
        b=b
if b>=a:
    print("No")
else:
    print("Yes")

```

Dry Run

Using any one of the inputs provided in the Exercise section above, dry run your psuedocode in the space below.

n=5	t= -2	
b=0	if -2 >=0: (false)	if b>=a: (false)
a=2	b+=1	print("No")
for i in range(1,6+1):	else:	else:
t= -1	b=b (b=1)	print("Yes")
if -1 >=0: (false)		
b+=1	t= -1	Output: Yes
else:	if -1 >=0: (false)	
b=b (b=0)	b+=1	
	else:	
	b=b (b=1)	
t= 0		
if 0 >=0: (true)	t= -3	
b+=1 (b=1)	if -3 >=0: (false)	
else:	b+=1	
b=b	else:	
	b=b (b=1)	

4. Beautiful Days

Lila goes to the market only on beautiful days. Given a range of numbered days starting from s and ending at e (where s and e are both included in the range) and a number k , determine the number of days in the range that are beautiful. Beautiful numbers are defined as numbers where $n - \text{reverse}(n)$ is evenly divisible by k . If a day's value is a beautiful number, it is a beautiful day.

Given a range, find the number of days Lila will go to the market.

Constraints

- $11 \leq s, e \leq 31$
- $0 \leq k \leq 10$

Interaction

The input comprises a single line containing 3 space-separated integers denoting the values of s , e and k respectively.

The output must contain a single number denoting the total number of days Lila goes to the market.

Sample

Input	Output
20 22 6	2
12 15 3	4

In the first case, the range of days is from 20 to 22 inclusive. The reverse of 20 is 02 or 2, reverse of 21 is 12 and reverse of 22 is 22. So $(20 - 2) \bmod 6 = 0$ and $(21 - 12) \bmod 6 = 3$ and $(22 - 22) \bmod 6 = 0$. Which means there are only two beautiful days within the range of 20 to 22. Lila will go to the market on these 2 days only.

In the second case, the range is from 12 to 15 inclusive. Reverse numbers of 12, 13, 14, 15 are 21, 31, 41, 51 respectively. $(12 - 21) \bmod 3 = 0$ and $(13 - 31) \bmod 3 = 0$ and $(14 - 41) \bmod 3 = 0$ and $(15 - 51) \bmod 3 = 0$. Which means all 4 days are beautiful therefore, Lila will go to the market for 4 days.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
12 14 4	0
27 31 5	1
17 19 6	2

Pseudocode

```

Input: s, e, k
Output:
for i in range(s,e+1):
    i=str(i)
    char=i[::-1]
    c=int(char)
    i=int(i)
    a=(i-c)%k
    if a==0:
        b=b+1
    else:
        b=b
print(b)

```

Dry Run

Using any one of the inputs provided in the Exercise section above, dry run your pseudocode in the space below.

```

Input: 12 14 4
for i in range(12,14+1):
    i=str(i)
    char=i[::-1] (char=21)
    c=int(char)
    i=int(i)
    a=(12-21)%4
    if a==0: (false)
        b=b+1
    else:
        b=b (b=0)

```

```

char=i[::-1] (char=31)
c=int(char)
i=int(i)
a=(13-31)%4
if a==0: (false)
    b=b+1
else:
    b=b (b=0)
char=i[::-1] (char=41)
c=int(char)
i=int(i)
a=(14-41)%4
if a==0: (false)
    b=b+1
else:
    b=b (b=0)
print(b)
Output: 0

```

5. Baby's First Question

You've just solved your first question on HackerRank. The question you solved has N test cases. You will receive a score according to the following conditions:

- If the solution passes all N test cases, you get 100 points.
- If it does not pass all test cases but passes the first M ($M < N$) test cases, you get K ($K < 100$) points.
- If neither of the above conditions are satisfied, you receive 0 points.

You are given a sequence of integers ($A_1 A_2 \dots A_N$) as a string S , where $A_i = 1$ if you passed the i th test case and $A_i = 0$ otherwise. In addition, you are also given the values of M and K . Determine how many points you will receive.

Constraints

- $N, M, K, A_1, A_2, \dots, A_N \in \mathbb{Z}$
- $1 \leq M < N \leq 10$
- $1 \leq K < 100$
- $A_i \in \{0, 1\}$

Interaction

The first line of the input contains 3 space-separated integers denoting the values of N , M , and K respectively. The second line of the input contains *string* S which is an N -long sequence of integers in the form $A_1 A_2 \dots A_N$.

The output must be a single integer denoting how many points you received.

Sample

Input	Output
4 2 50 "1011"	0
3 2 66 "110"	66

In the first case, $(N, M, K) = (4, 2, 50)$. Your solution does not pass all 4 test cases, nor does it pass the first 2 test cases, failing at test case 2. You receive 0 points.

In the second case, $(N, M, K) = (3, 2, 66)$. Your solution does not pass all 3 test cases, but it does pass the first 2 test cases. Hence, you receive 66 points.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
4 2 50 "1111"	100
5 3 30 "11011"	0
10 7 60 "1111101111"	0

Pseudocode

Input: N,M,K

Output:

n=4

m=2

k=50

count=0

s="1111"

for i in s:

```

if count==n:
    print(100)
elif count>=m:
    print(k)
else:
    print(0)

```

i=int(i)

if i==1 and i<=m:

count+=1

elif i>m:

count+=1

else:

count+=0

break

Dry Run

Using any one of the inputs provided in the Exercise section above, dry run your pseudocode in the space below.

Input: 5, 3, 30

s="11011"

for i in s:

i=int(i)

if i==1 and i<=m: (true)

count+=1 (count=1)

if i==1 and i<=m: (true)

count+=1 (count=2)

if i==1 and i<=m: (false)

count+=1

elif i>m: (false)

count+=1

else:

count+=0 (count=2)

break

#break statement breaks the loop

if count==n: (false)

print(100)

elif count>=m: (false)

print(k)

else:

print(0)

Output: 0

6. By Three They Come

A big phone service provider has just announced they will be handing out special prizes to some people registered with their service. The lucky winners will be selected on the basis of their phone number. A winning phone number must contain at least 5 integers which are a multiple of 3.

Given a numeric string S , determine if the string is a winning phone number.

Constraints

- S consists of only integers between 0 and 9
- $\text{len}(S) = 11$
- 0 is not considered a multiple of 3

Interaction

The first line of the input contains a numeric string S .

The output must be "YES" if the string is a winning phone number, and "NO" otherwise.

Sample

Input	Output
"09820561379"	"NO"
"03768391549"	"YES"

In the first case, the string $S = \text{"09820561379"}$ contains 4 multiples of 3; hence, it is not a winning phone number.

In the second case, the string $S = \text{"03768391549"}$ contains 5 multiples of 3; hence, it is a winning phone number.

Proposed Solution

```

1  count = 0
2  for i in S:
3      if int(i)%3 == 0: # convert the digit to integer, and check if divisible by 3
4          count += 1
5  if count > 5:
6      print('YES')
7  else:
8      print('NO')
```

Dry Run

Using the inputs provided in the Sample section above, dry run the proposed code solution below.

Input: 09820561379

count=0
for i in S:

if int(i)%3==0: (true) (i=0)
count+=1
if int(i)%3==0: (true)(i=9)
count+=1
if int(i)%3==0: (false)(i=8)
count+=1
if int(i)%3==0: (false)(i=2)
count+=1
if int(i)%3==0: (true)(i=0)
count+=1

if int(i)%3==0: (false)(i=5)
count+=1
if int(i)%3==0: (true)(i=6)
count+=1
if int(i)%3==0: (false)(i=1)
count+=1
if int(i)%3==0: (true)(i=3)
count+=1
if int(i)%3==0: (false)(i=7)
count+=1
if int(i)%3==0: (true)(i=9)
count+=1

if count>5: (true)
print("YES")
else:
print("NO")

Output: YES
(expected output= NO)

Error Identification

Briefly explain the errors you identified in the proposed code solution. Mention the line number and the errors in each line.

as the conditions says that zero should not be considered as the multiple of 3, in this code there was no condition to specify that situation that is why the answer is wrong.

Correct Solution

Rewrite the lines of code you mentioned above with their errors corrected.

```
Input: S
Output:
count=0
for i in S:
    i=int(i)
    if i==0:
        count+=0
    elif i%3==0:
        count+=1
    else:
        count=count
    if count>=5:
        print("YES")
    else:
        print("NO")
```

7. Introduction to Hackerrank

The guide to login/sign-up on Hackerrank tests is available here.

Problems are available on this link as a short introduction to programming:

<https://hr.gs/cs101-pw4-fall24>

Rough Work